

Wildfire Effects on AQI: What We Learned and How to Use It

Group 3

2026-03-09

Client: Air Quality Intelligence Platform

Stakeholder: Ontario municipal environmental offices, the Ontario Ministry of the Environment

Motivation:

- Explaining extreme AQI events: Tests whether wildfire activity explains AQI spikes missed by baseline models.
- Improving forecast reliability: Evaluates whether wildfire features improve next-day AQI predictions, especially on extreme days.
- Operational risk signal: Develops a wildfire-aware indicator to support early warnings and air-quality advisories.

Case Study 1 Recap

- Case 1 models performed well on average in predicting next-day AQI using traffic, meteorological, and temporal features. Linear models delivered the best overall performance, but all models struggled to capture episodic spikes in extreme AQI.
- The days when our baselines fail are the exact days our users rely on us most for health advisories.
- These aligned error spikes are driven by unobserved exogenous factors—specifically, wildfire smoke transport—rather than a lack of model complexity.

The key business question is: How can we make next-day AQI forecasts more reliable during high-risk pollution events, particularly during wildfire smoke episodes?

More specifically:

- Can wildfire activity explain extreme AQI spikes that current models fail to capture?
- Can wildfire signals be integrated into the forecasting pipeline to improve advisory reliability?

Data Processing Pipeline

Base Dataset:

- Case Study 1 station-day AQI dataset
- CWFIS Fire M3 Daily Hotspots archive (2022–2024)

Data Alignment:

- Match weather stations with nearby air quality stations (10 km radius)
- Convert hotspots to station-local days; Generate exposure lags (0–3 days)
- Merge wildfire features with AQI data;
- Handle missing data by dropping rows with missing targets and using forward fill or station-level median, producing model-ready observations.
- Fit the Imputer only on the training data and then transform both training and testing data to avoid data leakage

Wildfire Smoke Exposure Index

$$WSEI_{s,t} = \sum_{f \in F_{t-k}} \log(1 + I_f) \cdot K(d_{s,f}) \cdot W(\Delta\theta_{s,f}, v_s)$$

where

- I_f : Wildfire intensity (HFI / FRP / TFC)
- $K(d_{s,f})$: Distance-decay kernel
- $W(\Delta\theta_{s,f}, v_s)$: Weight hotspots by wind alignment and scale by wind speed
- $k = 0, 1, 2, 3$ day lags

Research Question 1

After controlling for meteorology, seasonality, and station effects, how strongly is wildfire activity associated with next-day AQI, and over what short lags (0–3 days)?

Modeling Approach

- Predict next-day AQI while isolating wildfire effects
- Incorporate lagged wildfire exposure (0–3 days) to capture smoke transport
- Control for meteorology, traffic, and calendar structure
- Hierarchical structure models station-specific baseline AQI levels
- Bayesian estimation provides uncertainty for wildfire impact

Fitting

- Posterior inference via MCMC (NUTS sampler)
- 4 chains $\times$ 2,000 iterations (per chain)

Posterior evidence

- The posterior probability that the cumulative 0–3 day wildfire effect is positive is $> 99.9\%$

MCMC diagnostics

- $\hat{R} \approx 1.00$ across parameters (good mixing / convergence)
- High effective sample sizes (ESS), suggesting estimates are stable and not driven by sampling noise

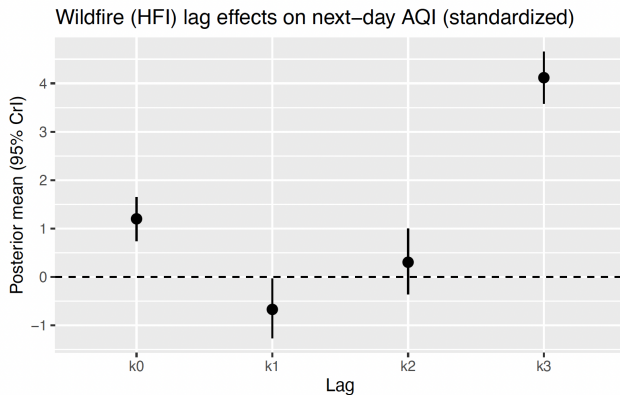
Wildfire Lag Effect

Lag-specific Wildfire Effect:

Wildfire exposure is associated with higher next-day AQI, with effects concentrated in short lags (0–3 days).

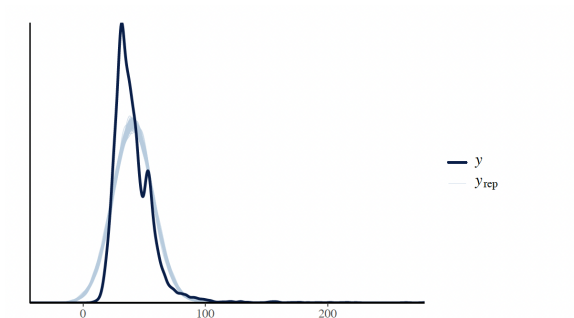
Cumulative Wildfire Effect:

- Mean ≈ 4.95 AQI increase
- 95% credible interval $\approx [4.63, 5.27]$
- Probability positive ≈ 1



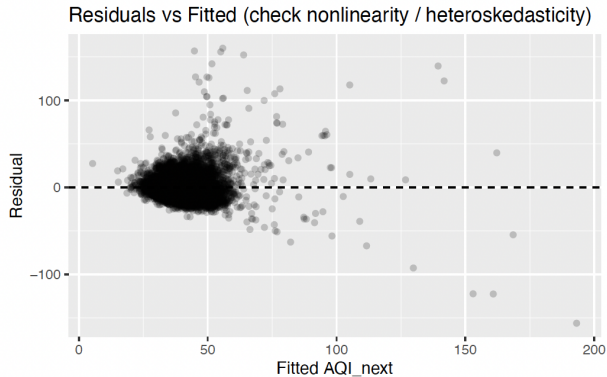
Posterior Predictive Check

- Model reproduces the main AQI distribution.
- Some mismatch in extreme right tail.
- Indicates difficulty modeling extreme events.



Residual Diagnostics

- Residual variance increases at higher AQI levels.
- Suggests heteroskedasticity during severe pollution days.



Heavy-Tail Modeling

- Student- t likelihood used to capture extreme pollution spikes.
- Estimated degrees of freedom ≈ 3.9 , indicating heavy-tailed AQI distribution.

Key takeaway:

- Strongest wildfire association appears at lag 3, with a clearly positive 95% credible interval. This likely reflects delayed smoke transport and accumulation in downwind regions before impacting the local AQI.

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	37.77	0.97	35.65	39.48	1.00	903	1596
wsei_hfi_k0_z	0.41	0.33	-0.21	1.08	1.00	5332	4688
wsei_hfi_k1_z	1.39	0.47	0.48	2.33	1.00	4393	4433
wsei_hfi_k2_z	-1.49	0.89	-3.20	0.14	1.00	3594	4321
wsei_hfi_k3_z	5.25	0.50	4.26	6.22	1.00	5884	4719
MeanTemp_z	0.64	0.82	-0.98	2.26	1.00	5597	4048
TotalPrecip_z	-1.66	0.12	-1.91	-1.42	1.00	8382	4630
MaxTemp_z	1.03	0.82	-0.57	2.64	1.00	4882	4461
MinTemp_z	0.37	0.77	-1.17	1.88	1.00	4035	4210
TempRange_z	1.84	0.30	1.25	2.43	1.00	4206	4131
TotalTraffic_z	-0.66	0.31	-1.27	-0.08	1.00	4906	4315
wind_dir_deg_z	-1.10	0.12	-1.34	-0.86	1.00	7462	4027

Gaussian vs Student- t

- Gaussian: thin tails $\rightarrow$ extremes inflate σ and can distort uncertainty.
- Student- t : heavy tails (small ν) $\rightarrow$ handles spikes without over-inflating σ .
- Robustness: wildfire lag effects (especially lag 3) stay positive in both models; Student- t is often cleaner/stronger because outliers have less leverage.

Remaining Limitations

- Wildfire hotspots are only proxies for smoke transport.
- Wind direction treated linearly.
- Lag coefficients may trade off due to correlation.
- Extreme AQI days remain hard to predict.

Research Question 2

Does adding wildfire exposure features improve next-day AQI point-forecast accuracy, with emphasis on extreme days (e.g., top decile AQI) where Case 1 failed?

Model Comparison Framework

We compare Baseline vs +Wildfire across three model families:

- OLS
- Neural Network (MLP)
- LSTM

Model Comparison Results

Overall performance comparison

	model_family		spec	RMSE	MAE
0	LSTM	+ Wildfire (cfg_2)		11.675581	8.716072
1	OLS	+ Wildfire		11.704901	8.697979
2	OLS	Baseline		11.726270	8.682664
3	LSTM	Baseline (cfg_2)		11.786356	8.939309
4	NN	Baseline		12.300621	9.295328
5	NN	+ Wildfire		12.782537	9.542219

Extreme-day performance comparison

	model_family		spec	top_decile_RMSE	top_decile_MAE
0	NN	Baseline		21.523922	18.380774
1	OLS	Baseline		22.470814	19.058420
2	NN	+ Wildfire		22.558778	19.144850
3	OLS	+ Wildfire		22.638192	19.316394
4	LSTM	Baseline (cfg_2)		23.453293	20.651726
5	LSTM	+ Wildfire (cfg_2)		23.531045	20.583334

- Overall: OLS $\approx$ LSTM; NN worse.
- Extreme days: NN baseline best; +Wildfire slightly worse (all models).
- Business: no strong smoke-day lift; avoid over-claim.
- Action: deploy OLS baseline; wildfire for monitoring/explanation.
- Next: transport-aware features; extreme-focused training.

Research Question 3

Can we produce a stable, interpretable wildfire-risk signal (feature + diagnostics) that supports stakeholder-facing explanations and monitoring?

Define the Operational Signal

- Operational target: “High AQI tomorrow” event.
- Definition: next-day AQI in the top 10% (binary outcome).
- Why this helps stakeholders: a single number for alerts + dashboards.

Step 1: Define the Monitored Outcome

Create a binary “high-AQI tomorrow” event (top 10% of next-day AQI).

Step 2: Fit an Interpretable Risk Model

Train a Bayesian hierarchical logistic regression.

The model’s wildfire exposure features, especially the 3-day lag, meaningfully increase the probability of a high-AQI day tomorrow.

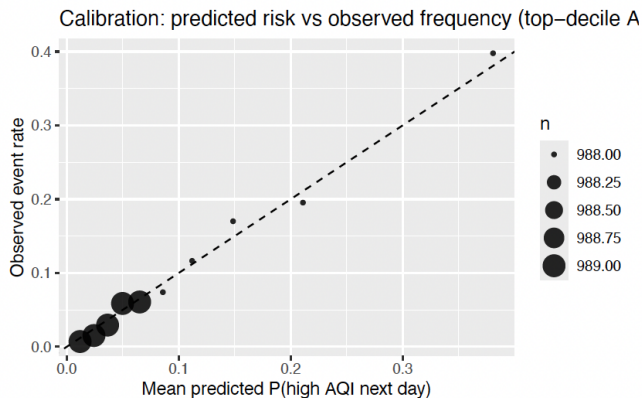
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-2.55	0.15	-2.86	-2.25	1.00	879	1600
wsei_hfi_k0_z	-0.07	0.04	-0.16	0.01	1.00	3291	2997
wsei_hfi_k1_z	0.17	0.06	0.05	0.29	1.00	2653	2630
wsei_hfi_k2_z	-0.17	0.07	-0.31	-0.03	1.00	2522	2574
wsei_hfi_k3_z	0.45	0.07	0.32	0.59	1.00	3234	2895
MeanTemp_z	0.19	0.82	-1.45	1.75	1.00	3065	2535
TotalPrecip_z	-0.43	0.06	-0.55	-0.33	1.00	4545	3039
MaxTemp_z	0.34	0.84	-1.36	1.95	1.00	2661	2558
MinTemp_z	0.18	0.77	-1.34	1.68	1.00	2204	2462
TempRange_z	0.41	0.28	-0.15	0.97	1.00	2173	2105
TotalTraffic_z	-0.14	0.07	-0.29	0.01	1.00	2337	2352
wind_dir_deg_z	-0.26	0.04	-0.33	-0.19	1.00	4419	2527

Result: Wildfire Increases High-AQI Risk

- Focus on the most actionable term: lag-3 wildfire exposure (k_3).
- Coefficient (log-odds): ≈ 0.45 .
- Odds ratio: $\exp(0.45) \approx 1.57$.
- $\rightarrow \sim 57\%$ higher odds of a high-AQI day per +1 SD wildfire exposure.
- Interpretation: strongest risk signal appears with ~ 3 -day delay.

Diagnostic A: Calibration of the Risk Score

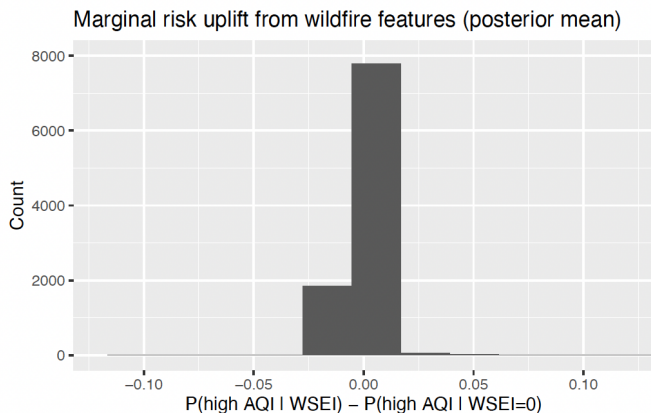
- Bin predictions into deciles of probability of high AQI next day.
- Near-diagonal points → probabilities are reasonably calibrated.



Diagnostic B: Wildfire-driven Risk Uplift

$$\Delta = P(\text{High AQI} \mid \text{WSEI}) - P(\text{High AQI} \mid \text{WSEI} = 0)$$

- mass near 0 $\rightarrow$ wildfire has little effect on most days (good: fewer false alarms)
- positive tail $\rightarrow$ wildfire meaningfully increases risk on smoke-impacted days



- Wildfire exposure is positively associated with next-day AQI, with the strongest effect at about a 3-day lag; results are robust under both Gaussian and Student- t errors.
- Adding WSEI improves performance during high-AQI periods by explaining spike days that baseline models tend to miss.
- The Bayesian risk model provides an interpretable probability score for top-decile AQI events; wildfire features, especially lag-3, raise predicted risk mainly on a limited subset of days.

Value to decision-making

- We can deploy a wildfire-aware early-warning indicator for Ontario: a daily risk probability + a clear explanation of why risk is elevated.